3.1 不定积分_2

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2. 若 $f(x)$ 的一个原函数是 e^{-x}，则 $\int f(x)dx =$ _____，$\int f'(x)dx =$ _____，$\int e^x f'(x)dx$ $=$ _____.

$F(x) = e^{-x}$ ， $f(x) = -e^{-x}$ $f'(x) = e^{-x}$

$$\int f(x)\,dx = \int -e^{-x}\,dx =$$

$$= e^{-x} + C$$

$$\int e^{-x}\,dx = -e^{-x} + C$$

$$\int e^{x} \cdot e^{-x}\,dx = \int dx = x + C$$

7. 不定积分 $\int \frac{(1-y)^2}{\sqrt[3]{y}}dy =$ _____.

$$= \int \frac{y^2+1-2y}{y^{\frac{1}{3}}}\,dy = \int \left(y^{2-\frac{1}{3}} + y^{-\frac{1}{3}} - 2y^{1-\frac{1}{3}}\right) dy$$

$$= \frac{1}{\frac{5}{3}+1} y^{\frac{5}{3}+1} + \frac{1}{-\frac{1}{3}+1} y^{-\frac{1}{3}+1} - 2\frac{1}{\frac{2}{3}+1} y^{\frac{2}{3}+1}$$

$$+ C$$

9. d _____ $= \frac{1}{\sqrt{1-x^2}}dx.$

y $\quad y'$

$dy = y'dx$

$$\int \frac{1}{\sqrt{1-x^2}}dx = \underbrace{\arcsin x + C}_{y}$$

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11. 不定积分$\int \frac{(1+\ln x)^{520}}{x}dx =$ _____.

$(\ln x)' = \frac{1}{x}$

$$\int (1+\ln x)^{520} d(\ln x+1)$$

$$= \frac{1}{520+1}(\ln x+1)^{520+1} + C.$$

$$\int x^k dx = \frac{1}{k+1}x^{k+1} + C$$

$$\int \square^{520} d\square = \frac{1}{520+1}\square^{520+1} + C$$

12. 不定积分$\int \frac{1}{1+\sin x}dx =$ _____.

$\sin 2x = 2\sin x\cos x$

方法一：$\int \frac{1}{1+2\sin\frac{x}{2}\cos\frac{x}{2}}dx$

$\sin x = \sin 2\frac{x}{2} = \underline{2\sin\frac{x}{2}\cos\frac{x}{2}}$

$\cos^2\frac{x}{2}$

同除$\cos^2\frac{x}{2}$

$= \int \frac{\sec^2\frac{x}{2}}{\sec^2\frac{x}{2}+2\tan\frac{x}{2}}dx$

$\sin x \quad \cos x$

$\Downarrow$

$\tan x \quad \sec^2 x$

$= 2\int \frac{d\tan\frac{x}{2}}{\tan^2\frac{x}{2}+2\tan\frac{x}{2}+1}$

$\sec^2 x - 1 = \tan^2 x$

$= 2\int \frac{1}{(\tan\frac{x}{2}+1)^2}d(\tan\frac{x}{2}+1)$

$\int \frac{1}{\square^2}d\square$

$= -2\frac{1}{1+\tan\frac{x}{2}} + C$

$= -\frac{1}{\square} + C$

补充：$\int \frac{1}{2+\cos x}dx$

$\frac{1+\cos 2\frac{x}{2}}{2} = \cos^2\frac{x}{2}$

$= \int \frac{1}{1+1+\cos x}dx$

$1+\cos x = 2\cos^2\frac{x}{2}$

$= \int \frac{1}{1+2\cos^2\frac{x}{2}}dx$

$= \int \frac{\sec^2\frac{x}{2}}{\sec^2\frac{x}{2}+2}dx$

$\int \frac{1}{a^2+x^2}dx$

$= \frac{1}{a}\arctan\frac{x}{a} + C$

$= 2\int \frac{1}{\quad}d\boxed{\tan\frac{x}{2}}$

$$= 2\int \frac{1}{3+\tan^2\frac{x}{2}}\, d\boxed{\tan\frac{x}{2}}$$

$$= \frac{1}{a}\arctan\frac{\square}{a} + C$$

$$\int \frac{1}{a^2+\square^2}\, d\square$$

$$= 2\cdot\frac{1}{\sqrt{3}}\arctan\left(\frac{\tan\frac{x}{2}}{\sqrt{3}}\right)+C$$

$$= \frac{1}{a}\arctan\frac{\square}{a}+C$$

方法二：

$$\int \frac{1}{1+\sin x}\,dx = \int \frac{1-\sin x}{(1+\sin x)(1-\sin x)}\,dx$$

$$= \int \frac{1}{\cos^2 x}\,dx - \int \frac{\sin x}{\cos^2 x}\,dx$$

13. 不定积分$\int\left(\frac{\sec x}{1+\tan x}\right)^2 dx =$ _____.

$$= \int \frac{\sec^2 x}{(1+\tan x)^2}\,dx$$

$$\int \frac{1}{x^2}\,dx = -\frac{1}{x}+C$$

$$= \int \frac{1}{(1+\tan x)^2}\,d(\tan x+1)$$

$$\int \frac{1}{\square^2}\,d\square = -\frac{1}{\square}+C$$

$$= -\frac{1}{1+\tan x}+C$$

14. 不定积分$\int 3x^2\sqrt[3]{1+x^3}\,dx=$ _____.

$$=\int \sqrt[3]{1+x^3}\,dx^3$$

$$\int \square^{\frac{1}{3}}\,d\square=\frac{1}{\frac{1}{3}+1}\square^{\frac{1}{3}+1}+C$$

$$=\int (1+x^3)^{\frac{1}{3}}\,d(x^3+1)$$

$$=\frac{1}{\frac{1}{3}+1}(1+x^3)^{\frac{1}{3}+1}+C$$

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15. 不定积分$\int xf'(ax^2+b)\,dx=$ _____. $(a\neq 0)$

$$=\frac{1}{2}\frac{1}{a}\int f'(\boxed{ax^2+b})\,d(ax^2+b)$$

$$\int f'(\square)\,d\square$$

$$=\frac{1}{2a}f(ax^2+b)+C$$

$$=f(\square)+C$$

16. 不定积分$\int \frac{f'(x)}{\sqrt{1-[f(x)]^2}}dx=$ _____.

16. 不定积分 $\int \frac{\quad}{\sqrt{1-[f(x)]^2}}\mathrm{d}x =$ _____.

$$= \int \frac{1}{\sqrt{1-f^2(x)}}\, d f(x)$$

$$= \arcsin f(x) + C$$

$$\int \frac{1}{\sqrt{1-\square^2}}\, d\square$$

$$= \arcsin \square + C$$

17. 不定积分 $\int \frac{\sqrt{1+2\arctan x}}{1+x^2}\mathrm{d}x =$ _____.

$$= \frac{1}{2}\int \sqrt{\boxed{1+2\arctan x}}\, d(2\arctan x+1)$$

$$= \frac{1}{2}\,\frac{2}{3}\,(1+2\arctan x)^{\frac{3}{2}} + C$$

$$\int \sqrt{\square}\, d\square$$

$$= \frac{2}{3}\,\square^{\frac{3}{2}} + C$$

18. 不定积分 $\int \frac{1}{\sqrt{3+2x}+\sqrt{2x-1}}\mathrm{d}x =$ _____.

$$= \int \frac{\sqrt{3+2x}-\sqrt{2x-1}}{(\sqrt{3+2x}+\sqrt{2x-1})(\sqrt{3+2x}-\sqrt{2x-1})}\, dx$$

$$\int \left(\sqrt{3+2x}+\sqrt{2x-1}\right)\left(\sqrt{3+2x}-\sqrt{2x-1}\right)$$

$$(a + b)\quad (a - b)$$

$$= \int \frac{\sqrt{3+2x}-\sqrt{2x-1}}{3+2x-(2x-1)}\,dx$$

$$= \int \left[\frac{1}{4}(3+2x)^{\frac{1}{2}} - \frac{1}{4}(2x-1)^{\frac{1}{2}}\right]dx$$

$$= \frac{1}{2}\frac{1}{4}\int (3+2x)^{\frac{1}{2}}\,d(2x+3) - \frac{1}{4}\cdot\frac{1}{2}\int (2x-1)^{\frac{1}{2}}\,d(2x-1)$$

$$= \frac{1}{8}\,\frac{1}{\frac{1}{2}+1}(3+2x)^{\frac{1}{2}+1} - \frac{1}{8}\,\frac{1}{\frac{1}{2}+1}(2x-1)^{\frac{1}{2}+1} + C$$

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19. 不定积分$\int \frac{1}{x^3}\sin\frac{1}{x}\mathrm{d}x =$ _____.

$$= \int \frac{1}{x}\cdot\boxed{\frac{1}{x^2}}\sin\frac{1}{x}\,dx$$

$$= -\int \boxed{\frac{1}{x}}\sin\boxed{\frac{1}{x}}\,d\boxed{\frac{1}{x}}$$

$$= -\int \boxed{\frac{1}{x}} \sin\boxed{\frac{1}{x}}\, d\boxed{\frac{1}{x}}$$

$$\overset{\frac{1}{x}=t}{=} -\int t \sin t \, dt$$

$$= \int t \, d\cos t$$

$$= t\cdot\cos t - \int \cos t \, dt$$

$$= t\cdot\cos t - \sin t + C$$

$$= \frac{1}{x}\cos\frac{1}{x} - \sin\frac{1}{x} + C$$

补充：

$$\int \frac{1}{x^3} e^{\frac{1}{x}} dx$$

$$= \int \frac{1}{x} \frac{1}{x^2} e^{\frac{1}{x}} dx$$

$$= -\int \frac{1}{x} e^{\frac{1}{x}} d\frac{1}{x}$$

$$\overset{\frac{1}{x}=t}{=} -\int t e^t dt$$

$$= -(te^t - e^t) + C$$

$$\int \sqrt{x} e^{\sqrt{x}} d\sqrt{x}$$

$$\overset{\sqrt{x}=t}{=}$$

21. 不定积分 $\int \frac{x+1}{x^2-2x+5}dx =$ _____.

分子为一次多项式，

分母为二次质因式 $(\Delta<0)$

$$= \int \frac{\frac{1}{2}(2x-2)+2}{x^2-2x+5} dx$$

做法：将分子凑成分母导数加常数

$$= \frac{1}{2}\int \frac{2x-2}{\ldots} dx + 2\int \frac{1}{\ldots} dx$$

$$= \frac{1}{2}\int \frac{2x-2}{x^2-2x+5}dx + 2\int \frac{1}{x^2-2x+5}dx$$

分子为 1

分母为二次质因式 $(\Delta<0)$

做法：凑平方和

$$\int \frac{1}{a^2+x^2}dx = \frac{1}{a}\arctan\frac{x}{a} + C$$

$$= \frac{1}{2}\int \frac{1}{x^2-2x+5}d(x^2-2x+5) + 2\int \frac{1}{x^2-2x+5}dx$$

$$= \frac{1}{2}\ln\left[x^2-2x+5\right] + 2\int \frac{1}{(x-1)^2+2^2}dx$$

$$= \frac{1}{2}\ln(x^2-2x+5) + 2\cdot\frac{1}{2}\arctan\frac{x-1}{2} + C.$$

22. 不定积分 $\int \frac{1+2x^2}{x^2(1+x^2)}dx =$ _____.

分子能凑分母的因式.

$$= \int \frac{1+x^2+x^2}{x^2(1+x^2)}dx$$

$$= \int \frac{1+x^2}{x^2(1+x^2)}dx + \int \frac{x^2}{x^2(1+x^2)}dx$$

$$= \int \frac{1}{x^2}dx + \int \frac{1}{1+x^2}dx$$

$$= -\frac{1}{x} + \arctan x + C$$

25. 设 $f(x)=e^{-x}$，则 $\int xf'(x)dx=$ _____.

法一：$f(x)=e^{-x}$，$f'(x)=-e^{-x}$

则 $-\int xe^{-x}dx=\int x\,de^{-x}$

$=x\cdot e^{-x}-\int e^{-x}dx$

$=x\cdot e^{-x}+e^{-x}+C$

法二：$\int xf'(x)dx$

$=\int x\,df(x)$

$=xf(x)-\int f(x)dx$

$=xe^{-x}-\int e^{-x}dx$

$=xe^{-x}+e^{-x}+C$

26. 若函数 $f(x)$ 有连续的二阶导数，则不定积分 $\int xf''(x)dx=$ _____.

$\int xf''(x)dx=\int x\,df'(x)$

$=xf'(x)-\int f'(x)dx$

$=xf'(x)-f(x)+C$

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27. 不定积分$\int x\tan^2 x\mathrm{d}x =$ _____.

$$= \int x(\sec^2 x - 1)\,dx$$

$$= \int x\cdot\sec^2 x\,dx - \int x\,dx$$

$$= \int x\,d\tan x - \frac{1}{2}x^2$$

$$= x\tan x - \int \tan x\,dx - \frac{1}{2}x^2$$

$$= x\tan x - (-\ln|\cos x|) - \frac{1}{2}x^2 + C$$

$$= x\tan x + \ln|\cos x| - \frac{1}{2}x^2 + C$$

三、计算题

1. 求不定积分$\int \frac{x^3-27}{x-3}\mathrm{d}x$.

2. 计算不定积分$\int\left(\sqrt[3]{x}-\frac{1}{\sqrt{x}}\right)\left(\sqrt{x}+\frac{1}{\sqrt[3]{x}}\right)\mathrm{d}x$.

3. 计算不定积分$\int\frac{x^2}{\sqrt{2x^3-1}}\mathrm{d}x$.

4. 计算不定积分$\int\frac{\sin x-\cos x}{\sin x+\cos x}\mathrm{d}x$.

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5. 计算不定积分$\int\frac{\sin x\cos x}{1+\sin^4 x}\mathrm{d}x$.

6. 计算不定积分$\int \sin^3 x \mathrm{d}x$.

7. 计算不定积分$\int \sec^6 x \mathrm{d}x$.

8. 计算不定积分$\int \frac{\sec t}{2\sec^2 t - 1} \mathrm{d}t$.